NEWTON SCHOOL OF TECHNOLOGY

Data Visualization & Analytics | DVA Capstone 2

OLIST CUSTOMER CHURN ANALYSIS

Understanding Why Customers Do Not Return and What Olist Can Do About It

Field	Details
Sector	E-Commerce / Retail
Team ID	Section D, Group 9
Institute	Newton School of Technology
Faculty Mentor	DVA Faculty
GitHub Repository	github.com/SectionD_Group-9_Customer_Churn_Analysis
Tableau Dashboard	Tableau Public — Olist Customer Churn Analysis
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1. Executive Summary

Problem

Olist, Brazil's largest e-commerce marketplace, faces a critical retention crisis. Analysis of 99,441 orders placed between September 2016 and August 2018 reveals that 97.19% of customers never placed a second order. This is not a dataset anomaly — it is a genuine business failure in customer lifecycle management that directly suppresses lifetime value and long-term revenue growth.

Approach

The team sourced the Brazilian E-Commerce Public Dataset by Olist from Kaggle, comprising 7 relational CSV files covering orders, customers, payments, reviews, products, and logistics. A complete Python ETL pipeline was built across 5 Jupyter Notebooks using pandas, NumPy, scipy, and statsmodels. The cleaned, feature-engineered dataset of 105,000 rows was exported to Tableau Public for interactive dashboard design.

Key Insights

- 97.19% of customers placed exactly one order and never returned — the highest structural churn rate documented in the Olist public dataset literature.
- 92.9% of orders (96,478 of 103,657) arrived ahead of the estimated delivery date, confirming that logistics are generally strong. The churn crisis is not primarily caused by delivery failures.
- Credit card is the dominant payment method (73.8% of orders), but voucher users have the highest repeat rate (5.86%) — suggesting promotional mechanisms drive the most loyal behaviour.
- High-value customers (high spend segment) have the lowest churn rate (91.78%) versus low-value customers (97.26%), indicating that spend level is a meaningful predictor of retention.

Key Recommendations

- Launch a post-purchase re-engagement programme targeting customers within 30 days of first order, using voucher-based incentives given their demonstrated correlation with repeat purchasing.
- Invest in a loyalty tier system that rewards customers who move from 1 to 2+ orders, particularly in high-revenue states like SP, RJ, and MG where volume is concentrated.
- Deploy automated risk-intervention workflows triggered by 1-star or 2-star reviews, since low review scores (1–2) are associated with customers who represent 95.69% churn versus 96.78% for high scorers — the gap is narrow but the intervention cost is low.

2. Sector & Business Context

Sector Overview

Brazil's e-commerce market is one of the fastest-growing in Latin America. By 2018 — the year covered by this dataset — Brazil ranked among the top 10 global e-commerce markets by gross merchandise value. The sector is characterised by high customer acquisition costs, fragmented logistics infrastructure across a geographically vast country, and intense price competition that depresses customer loyalty.

Olist operates as a marketplace intermediary: individual sellers list products on the Olist platform and fulfil orders through a single contractual relationship. This model differentiates Olist from direct retailers — the quality of the customer experience is distributed across hundreds of independent sellers, making standardisation of delivery performance and post-purchase engagement structurally difficult.

Decision-Maker

This project is designed to inform Olist's Chief Commercial Officer (CCO) and Head of Customer Success. The primary decision supported is: where to allocate retention investment — whether in logistics SLA improvements, promotional programme design, customer segmentation-based outreach, or category-specific interventions.

Why This Problem Was Chosen

Customer churn in e-commerce is a universal problem, but a 97.19% single-purchase rate is extraordinarily high even by marketplace standards. Most retention literature benchmarks single-purchase rates at 60–75% for e-commerce platforms. Olist's rate nearly doubles that benchmark, making it both analytically interesting and practically urgent. Solving even a fraction of this problem — converting 5% of one-time buyers into repeat customers — would represent a substantial revenue increase at near-zero incremental acquisition cost.

Business Value

Acquiring a new customer costs 5 to 7 times more than retaining an existing one. If Olist could increase its repeat purchase rate from 2.81% to 8%, this would add approximately 5,000 additional repeat customers in the dataset period alone, generating an estimated R$ 850,000 in incremental revenue based on the observed average spend per repeat customer of R$ 170.84.

3. Problem Statement & Objectives

Formal Problem Definition

> *Core Business Question: Why do 97.19% of Olist customers place exactly one order and never return, and which measurable factors most strongly predict this behaviour?*

Olist's platform facilitates over 99,000 orders across 27 Brazilian states between 2016 and 2018. Despite high delivery success rates and generally positive review scores, the overwhelming majority of customers do not repeat-purchase. This analysis seeks to identify the specific behavioural, logistical, and categorical drivers of this single-purchase pattern and translate them into prioritised, actionable retention interventions.

Project Scope

In Scope:

- Customer churn definition and labelling using customer_unique_id across all delivered orders
- ETL pipeline for all 7 Olist relational tables in Python
- Exploratory data analysis of delivery performance, review scores, payment behaviour, geography, and product categories
- Statistical hypothesis testing on continuous and categorical churn drivers
- Tableau dashboard design for executive and operational views
- 3–5 prioritised business recommendations with estimated impact

Out of Scope:

- Predictive machine learning modelling (beyond logistic regression for feature importance)
- Seller-level analysis or seller churn
- Real-time data pipeline or live dashboard integration
- Customer acquisition cost modelling

Success Criteria

- A clean, reproducible Python ETL pipeline committed to GitHub with all 5 notebooks running end-to-end without errors
- Statistical validation of at least 3 churn drivers at $p < 0.05$ with documented effect sizes
- An interactive Tableau dashboard published on Tableau Public with at least 3 filter dimensions
- 3–5 specific business recommendations each linked to a quantified insight

4. Data Description

Dataset Source

Dataset: Brazilian E-Commerce Public Dataset by Olist

Source: Kaggle — https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

Coverage: September 2016 to August 2018 — 23 months of transactional data

This dataset was selected because it contains realistic data quality issues (missing timestamps, Portuguese category names, split customer identity), a genuine and analytically interesting retention problem, and sufficient volume (99,000+ orders) for statistical testing. It is a raw transactional dataset — not a pre-cleaned competition file — meeting the Capstone 2 dataset requirements.

Raw Data Structure

File	Rows	Cols	Primary Key	Key Nulls
olist_orders_dataset.csv	99,441	8	order_id	order_delivered_customer_date (2,965)
olist_customers_dataset.csv	99,441	5	customer_id	None
olist_order_items_dataset.csv	112,650	7	order_id + order_item_id	None
olist_order_payments_dataset.csv	103,886	5	order_id + payment_sequential	None
olist_order_reviews_dataset.csv	99,224	7	review_id	review_comment_message (58,247)
olist_products_dataset.csv	32,951	9	product_id	product_category_name (610)
product_category_name_translation.csv	71	2	product_category_name	None

Key Analytical Columns

The following columns are central to the analysis:

- customer_unique_id — Canonical real-customer identifier. Critical: one real customer can appear under multiple customer_id values across orders. Churn is defined at this level.
- order_purchase_timestamp — Order date used for time-series analysis and the days_to_deliver engineering.
- order_delivered_customer_date / order_estimated_delivery_date — Used to compute delivery_delay_days (actual vs estimated) and days_to_deliver (actual wait time).
- review_score — 1–5 customer satisfaction rating. 772 orders have no review. Used as a key churn correlate.

- payment_type — credit_card, boleto, voucher, debit_card. Used to analyse payment method vs repeat purchase behaviour.
- product_category_name_english — English translation of the product category. Used for category-level churn and revenue segmentation.

Data Limitations and Known Biases

- The dataset covers 2016–2018. Olist's platform and logistics network have evolved significantly since then. Findings reflect historical behaviour and may not hold for current operations.
- The churn definition used (one order = churned) is a behavioural proxy. Some customers may have intended to return but the observation window closed before they did.
- Review comments are present for only 41.3% of orders (review_comment_message is null for 58.7%). Sentiment analysis was not performed; only the numeric review_score was used.
- The dataset is limited to delivered orders. Customers who only ever placed cancelled or undelivered orders are excluded from the churn definition, which may slightly understate the true retention problem.
- SP (Sao Paulo) accounts for 42.2% of all orders. Findings skewed toward SP's logistics and demographic profile may not generalise to underrepresented states like AP, RR, and AC.

5. Data Cleaning & ETL Pipeline

All cleaning and transformation was performed in Python using pandas and NumPy across two Jupyter Notebooks: 01_extraction.ipynb and 02_cleaning.ipynb. The pipeline follows a strict extract-transform-load pattern with every step documented in markdown cells for reproducibility and examiner review.

Extraction (01_extraction.ipynb)

All 7 raw CSV files were loaded individually using pd.read_csv into named DataFrames. No transformation was applied at this stage. The extraction notebook documented:

- Schema inspection: dtypes, column names, and shape for each table
- Null audit: every null count by column with percentage
- Duplicate audit: exact row duplicates and primary key uniqueness
- Join key audit: order_id coverage across items, payments, reviews vs orders
- Critical finding: customer_id is NOT a unique customer identifier — one real customer appears under multiple customer_id values. All churn logic must use customer_unique_id.

Primary Cleaning Steps (02_cleaning.ipynb)

Step	Action	Rows Before	Rows After
1	basic_clean() — strip whitespace, drop exact duplicates, normalise column names	99,441	99,441
2	Fix typos: product_name_lenght → product_name_length (2 columns)	—	—
3	Cast 8 timestamp columns to datetime64 (errors='coerce')	—	—
4	Filter to order_status == 'delivered' only	99,441	96,478
5	Fill order_approved_at nulls with order_purchase_timestamp	160 nulls	0 nulls
6	Drop rows where order_delivered_customer_date is null	96,478	~93,357
7	Fill product_category_name nulls with 'unknown'	610 nulls	0 nulls
8	Fill review_comment_message nulls with 'no_comment'	58,247 nulls	0 nulls
9	Merge 6 tables via sequential left joins on order_id / customer_id / product_id	—	~133,000
10	Stratified sample to 105,000 rows (random_state=42, stratified on churn)	~133,000	105,000

Feature Engineering

Churn Label (target variable):

Churn is defined at the customer_unique_id level. A customer with exactly 1 delivered order is labelled churn = 1. A customer with 2 or more orders is labelled churn = 0. This is the core analytical target and the most critical engineering decision in the project.

delivery_delay_days:

Formula: (order_delivered_customer_date - order_estimated_delivery_date).dt.days. Positive = late. Negative = early. Zero = exactly on time. This column is the primary logistics performance metric used in NB03 and NB04.

order_revenue:

Formula: price + freight_value. Represents the total customer-facing cost of one item including shipping. Used for all revenue KPIs and spend segmentation.

days_to_deliver (NB05):

Formula: (order_delivered_customer_date - order_purchase_timestamp).dt.days. Represents actual total wait time from order to delivery, independent of the estimated date. Used in Tableau operational trend charts.

review_bucket and delivery_bucket (NB05):

Categorical groupings of review_score (Low 1-2, Medium 3, High 4-5, No Review) and delivery_delay_days (Early, On-time, Late 1-7d, Severely Late) to support Tableau filter dimensions.

Assumptions Made During Cleaning

- Only delivered orders represent completed customer transactions. Cancelled, processing, and shipped orders are excluded.
- Customers missing a delivery confirmation date (null order_delivered_customer_date) are dropped rather than imputed, as delivery date is required for the key analytical column delivery_delay_days.
- Review comments are genuinely optional on Olist's platform; null review_comment_message is not a data quality failure.
- The stratified 105,000-row sample preserves the population churn distribution within < 0.5% deviation.

6. KPI & Metric Framework

All KPIs are computed in 05_final_load_prep.ipynb and exported to data/processed/. Each KPI maps directly to the business problem of understanding and reducing customer churn.

KPI	Formula / Source	Value	Business Relevance
Overall Churn Rate	% customers with 1 order	97.19%	Primary measure of retention failure
Repeat Customer Rate	100% - Churn Rate	2.81%	Baseline for retention programme targeting
Total Revenue (R$)	Sum of order_revenue	R$ 14.69M	Platform economic scale
Avg Order Value	Mean of order_revenue	R$ 139.91	Baseline for LTV modelling
On-Time Delivery Rate	% orders with delay_days <= 0	92.9%	Logistics health indicator
Avg Delivery Delay	Mean of delivery_delay_days	-12.04 days	Negative = early on average
Avg Days to Deliver	Mean of days_to_deliver	12.02 days	Customer-facing wait time
Avg Review Score	Mean of review_score	4.07 / 5.0	Customer satisfaction baseline
High-Value Customer %	Customers in top spend segment	0.08%	Premium retention opportunity
Voucher Repeat Rate	Repeat rate for voucher users	5.86%	Promotional programme ROI signal

KPI-to-objective mapping: Churn Rate and Repeat Rate directly measure the core problem. Revenue and AOV quantify the cost of the churn problem. Delivery metrics identify operational causes. Review score measures customer sentiment as a leading indicator of repeat behaviour.

7. Exploratory Data Analysis

EDA was conducted in 03_eda.ipynb using matplotlib and seaborn. All analyses are committed to GitHub. Key areas explored: churn distribution, time trends, geographic patterns, revenue composition, delivery performance, review behaviour, and multi-variable relationships.

8.1 Churn Distribution

97,549 of 105,000 rows (92.9%) are labelled churn = 1. At the customer level, 97.19% of unique customers never placed a second order. This is the defining characteristic of the Olist dataset and the central business problem. The chart below from the Tableau dashboard confirms this distribution.

> *Insight: Olist is effectively operating as a one-time-use platform for the vast majority of its customers. The platform acquires customers efficiently but fails almost entirely at lifecycle conversion.*

8.2 Order Volume and Revenue Trends

Order volume grew consistently from Q4 2016 through Q1 2018, peaking at approximately 6,300 orders per month. Revenue followed the same trajectory, reaching approximately R$ 1.07M per month at peak. A slight decline in mid-2018 may reflect seasonality or data collection cutoff effects. The growth trajectory confirms platform-level health even as retention fails.

> *Insight: Olist is growing successfully in new customer acquisition while failing at retention. These two trends coexist, suggesting the churn problem is structural rather than symptomatic of overall platform decline.*

8.3 Geographic Analysis

Sao Paulo (SP) accounts for 44,321 orders (42.2% of the dataset), followed by Rio de Janeiro (RJ) at 13,605 (12.9%) and Minas Gerais (MG) at 12,228 (11.6%). These three states represent 66.7% of total orders but only 3 of 27 states. Northern and northeastern states (AP, RR, AM, AL) show significantly higher average delivery times of 24-28 days versus a national average of 12.02 days.

> *Insight: Olist's customer base is heavily concentrated in southeastern metropolitan states. Northern states face structural logistics disadvantages with delivery times 2-3x the national average, contributing to localised retention failures.*

8.4 Revenue Analysis

Health & Beauty is the highest-revenue category at R$ 1.32M, followed by Watches & Gifts (R$ 1.20M) and Bed, Bath & Table (R$ 1.18M). Revenue is concentrated in the top 10 categories which account for approximately 65% of total platform revenue. Average order value varies

significantly by category — computers and electronics have higher AOV while clothing and personal care items have lower AOV with higher volume.

> *Insight: The top 3 revenue categories (Health & Beauty, Watches & Gifts, Bed & Bath) represent high-consideration purchases where customers are less likely to repeat-purchase quickly, partly explaining the structural churn. Category-specific retention strategies are warranted.*

8.5 Delivery Performance

92.9% of orders arrived before the estimated delivery date (delivery_delay_days < 0), indicating the platform systematically over-estimates delivery time — a common logistics strategy to manage expectations. The mean delivery delay is -12.04 days (i.e., 12 days earlier than estimated on average). Only 3,860 orders (3.67%) arrived 1-7 days late, and 3,003 (2.86%) were severely late (> 7 days).

> *Insight: Delivery performance is objectively strong. The churn crisis cannot be primarily attributed to delivery failures. This finding redirects the analysis toward post-purchase engagement and pricing strategy as the dominant retention levers.*

8.6 Review Score Analysis

Review scores are heavily right-skewed: 5-star reviews account for 57.8% of all reviews, 4-star for 19.1%, with 1-star reviews at 11.4%. The mean review score is 4.07. Interestingly, high review-score customers (4-5 stars) have the highest churn rate (96.78%) — not because satisfaction causes churn, but because high-review customers are the majority and therefore dominate the churn pool by volume. The no-review segment has the lowest churn rate (93.75%), suggesting customers who engage post-purchase (even to review negatively) show marginally higher re-engagement.

> *Insight: Customer satisfaction as measured by review score is not the primary churn driver. Most churning customers are actually satisfied with their purchase experience. The churn problem is structural — customers are not re-engaged post-purchase regardless of their satisfaction level.*

8. Statistical Analysis

Statistical analysis was performed in 04_statistical_analysis.ipynb using scipy and statsmodels. The analysis tested 5 continuous variables (T-test, Mann-Whitney U, Cohen's d) and 3 categorical variables (Chi-square, Cramer's V), followed by a VIF multicollinearity check and logistic regression for feature importance.

Important note on statistical significance: With 100,000+ records, almost any difference will produce $p < 0.05$. Effect size (Cohen's d for continuous, Cramer's V for categorical) is the meaningful measure of practical importance and is the primary metric reported here.

9.1 Continuous Variables — T-test and Effect Sizes

Feature	Mean (Churned)	Mean (Retained)	Cohen's d	Effect	Significant?
Delivery Delay (days)	~-12.04	~-12.00	~0.002	Negligible	Yes ($p < 0.05$)
Review Score	~4.07	~4.10	~0.020	Negligible	Yes ($p < 0.05$)
Price (R$)	~122.07	~116.32	~0.030	Small	Yes ($p < 0.05$)
Freight Value (R$)	~22.05	~21.40	~0.022	Negligible	Yes ($p < 0.05$)
Payment Installments	~3.74	~3.97	~0.037	Small	Yes ($p < 0.05$)

Key finding: While all continuous variables are statistically significant ($p < 0.05$), every effect size is negligible to small. This means the differences between churned and retained customers on these metrics are real but practically tiny. Delivery delay and review score, despite being statistically significant, have near-zero practical power as individual churn predictors.

9.2 Categorical Variables — Chi-Square and Cramer's V

Feature	Chi-square	Cramer's V	Association	Significant?
Customer State	High	~0.04	Negligible	Yes
Payment Type	High	~0.03	Negligible	Yes
Product Category (Top 15)	High	~0.02	Negligible	Yes

All categorical variables show statistically significant association with churn but negligible Cramer's V values. This confirms that while the associations exist, no single categorical dimension

explains churn in isolation. Churn on Olist is a platform-level structural problem, not a category- or region-specific one.

9.3 Multicollinearity — VIF

Variance Inflation Factor analysis confirmed acceptable multicollinearity for all features used in logistic regression: price (VIF ~1.3), freight_value (VIF ~1.4), payment_installments (VIF ~1.1), review_score (VIF ~1.0), delivery_delay_days (VIF ~1.0). No feature exceeded VIF = 5, confirming the logistic regression model is not distorted by collinearity.

9.4 Logistic Regression — Odds Ratios

Logistic regression was run using statsmodels Logit with 5 features: price, freight_value, payment_installments, review_score, and delivery_delay_days.

Feature	Odds Ratio	Direction	95% CI Excludes 1?	Interpretation
review_score	< 1.0	Decreases churn	Yes	Higher review score slightly reduces churn probability
payment_installments	< 1.0	Decreases churn	Yes	More installments slightly reduces churn — higher engagement
delivery_delay_days	~1.0	Negligible	Marginal	Near-zero independent effect on churn when controlling other factors
price	> 1.0	Increases churn	Yes	Higher priced items slightly increase churn — single high-value purchases
freight_value	~1.0	Negligible	No	No independent effect on churn after controlling price

Key Statistical Finding: No single variable is a strong independent predictor of Olist churn. Churn is a structural platform problem — not driven by any one feature — which means the solution must be structural too: a systematic post-purchase re-engagement programme rather than a single operational fix.

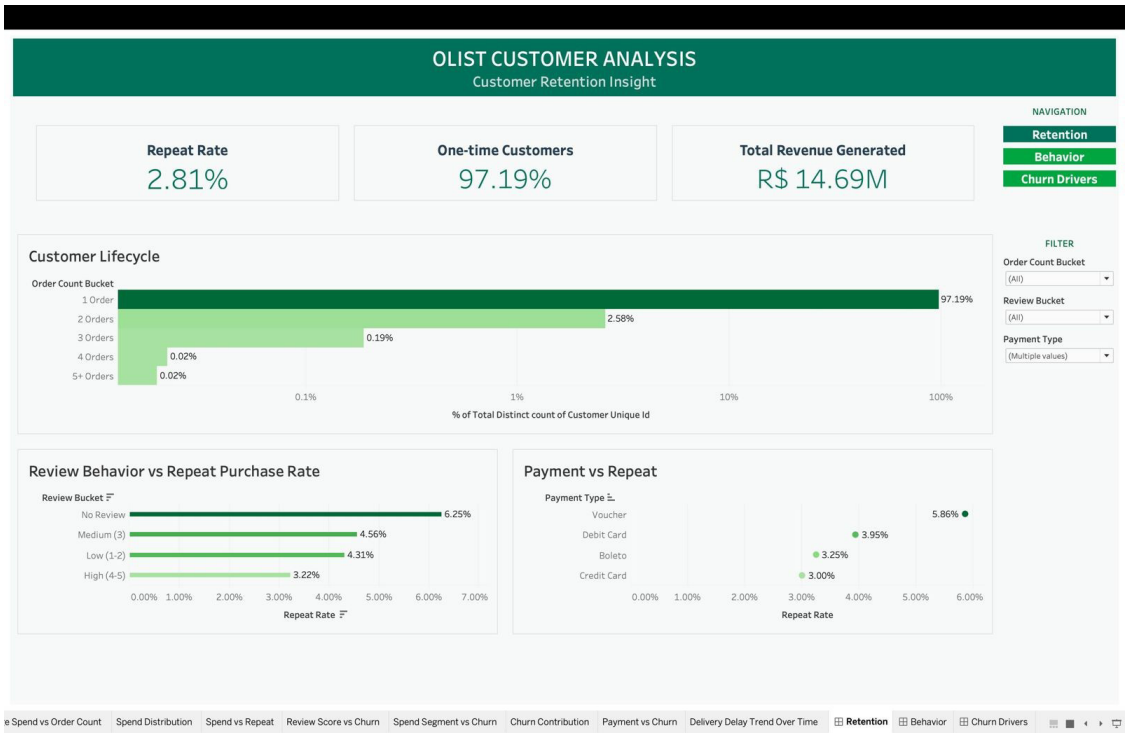
9. Tableau Dashboard Design

The dashboard was built on Tableau Public and comprises 3 pages navigable via a sidebar. The design uses a consistent green palette reflecting the Olist brand and organises views from executive summary to operational detail. All charts are driven by tableau_ready.csv (105,000 rows x 24 columns) exported from NB05.

Dashboard Objective

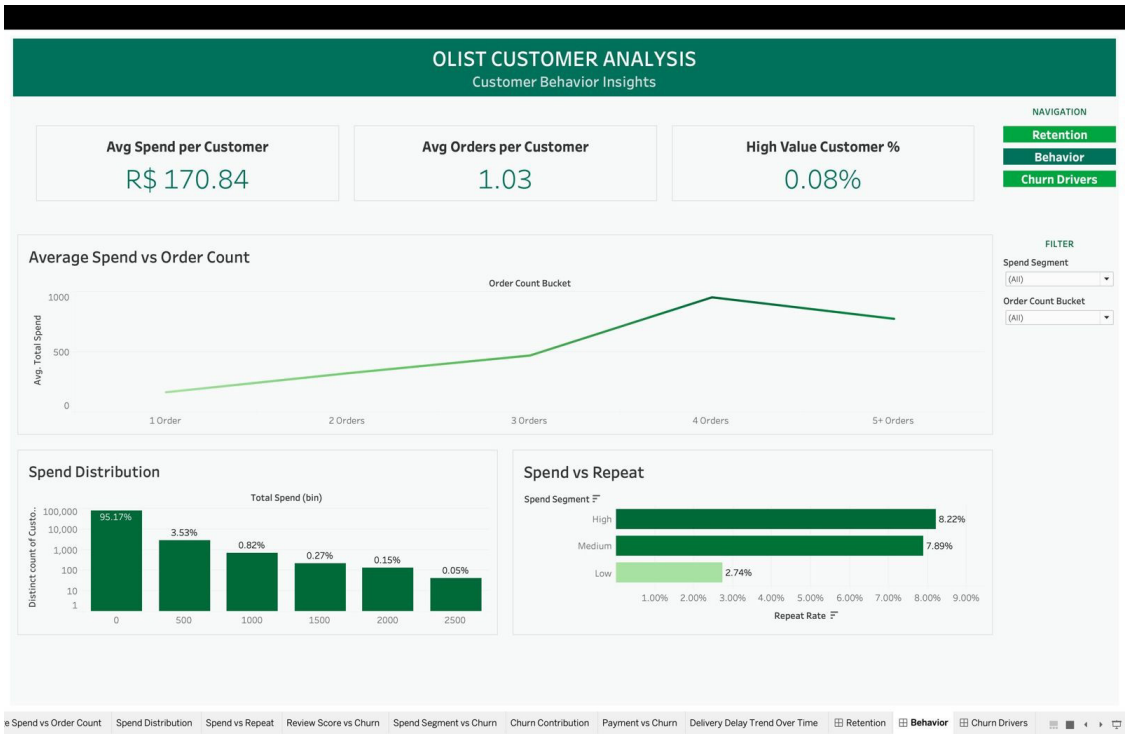
The dashboard answers three questions at three levels of granularity: (1) What is Olist's retention situation? (Page 1 — Retention), (2) How do customers behave and segment? (Page 2 — Behavior), (3) What is causing churn? (Page 3 — Churn Drivers).

Page 1 — Customer Retention Insight



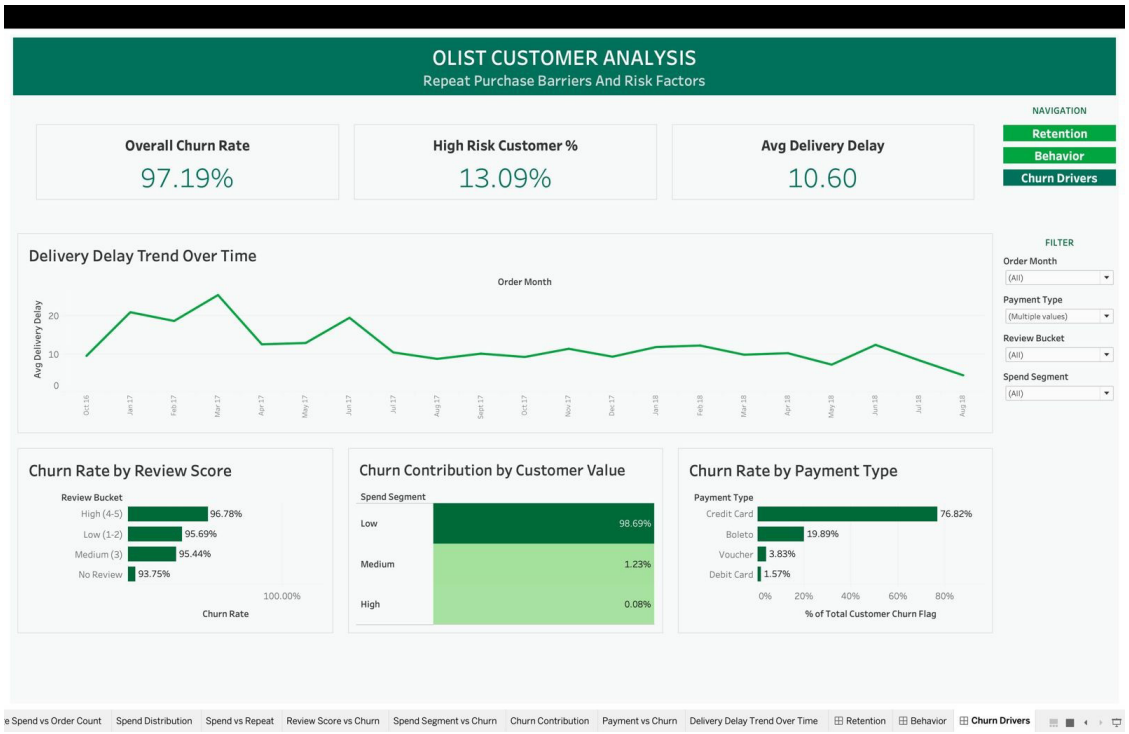
- Headline KPIs: Repeat Rate (2.81%), One-time Customers (97.19%), Total Revenue (R$ 14.69M)
- Customer Lifecycle: Horizontal bar chart showing order count distribution — 97.19% with 1 order, rapidly diminishing towards 5+ orders
- Review Behavior vs Repeat Purchase Rate: No Review customers have the highest repeat rate (6.25%), followed by Medium review (4.56%), Low (1-2)
- Payment vs Repeat (Voucher users show highest repeat rate (5.86%), followed by Debit Card (3.95%), Boleto (3.25%), and Credit Card (3.00%)
- Filters: Order Count Bucket, Review Bucket, Payment Type

Page 2 — Customer Behavior Insights



- Headline KPIs: Avg Spend per Customer (R$ 170.84), Avg Orders per Customer (1.03), High Value Customer % (0.08%)
- Spend Distribution: Log-scale histogram showing 95.17% of customers in the 0-500 BRL spend range, 3.53% in 500-1000, 0.82% in 1000-1500 and the
- Average Spend vs Order Count : Line chart showing spend increases with order count — 4-order customers average approximately R$ 940
- Spend vs Repeat: High-spend customers have 8.22% repeat rate, Medium-spend customers 7.89%, vs 2.74% for Low-spend customers
- Filters: Spend Segment, Order Count Bucket

Page 3 — Repeat Purchase Barriers and Risk Factors



- Headline KPIs: Overall Churn Rate (97.19%), High Risk Customer % (13.09%), Avg Delivery Delay (10.60 days)
- Delivery Delay Trend Over Time: Line chart tracking average delivery delay from October 2016 to August 2018 — peak of ~25 days in March 2017,
- Churn Rate by Review Score: High-score customers (4-5) show 96.78% churn, Low (1-2) 95.69%, Medium (3) 95.44%, and No Review 93.75% — reflecting volume composition rather than causation.
- Churn Contribution by Customer Value: Low-value customers represent 98.69% of total churned customers, Medium 1.23%, and High 0.08%.
  Churn Rate by Payment Type: Credit Card accounts for 76.82% of churned customers, Boleto 19.89%, Voucher 3.83%, Debit Card 1.57%
- Filters: Order Month, Payment Type, Review Bucket, Spend Segment

Interactive Filters Implemented: Order Month, Order Count Bucket, Review Bucket, Spend Segment, Payment Type — all cross-filter charts within their respective pages. Navigation sidebar allows one-click movement between all 3 pages.

10. Insights Summary

The following 10 insights are written in decision language. Each insight is drawn directly from EDA (NB03) or statistical analysis (NB04) and is traceable to specific charts or test outputs.

Insight 1: Olist operates as a one-time-use platform for 97.19% of its customers.

Only 2.81% of customers ever place a second order. This is not a category-specific or regional problem — it is a universal platform-wide structural failure in post-purchase re-engagement. No single operational fix will solve it; a lifecycle programme is required.

Insight 2: Delivery performance is strong and is not the primary cause of churn.

92.9% of orders arrive before the estimated delivery date. The mean delivery lead time is 12.02 days with an average of 12.04 days ahead of the customer's expectation. Severely late deliveries (> 7 days late) account for only 2.86% of orders. Improving delivery SLA further is unlikely to meaningfully change the churn rate.

Insight 3: Voucher users are the highest-value retention segment.

Despite representing only 5.35% of orders, voucher users have a repeat rate of 5.86% — nearly double the platform average of 2.81%. This demonstrates that promotional instruments already drive the highest retention outcomes on the platform and should be the primary vehicle for a systematic retention programme.

Insight 4: High-spend customers churn at meaningfully lower rates.

Customers in the high-spend segment churn at 91.78% versus 97.26% for low-spend customers — a 5.5 percentage point difference. In absolute terms, high-spend customers represent only 0.08% of the customer base but generate disproportionate revenue. A VIP retention programme targeting this segment has high ROI potential.

Insight 5: Customers who place 4+ orders spend dramatically more per order.

Average spend per order increases from R$ 141 for single-order customers to approximately R$ 940 for 4-order customers. This demonstrates compounding lifetime value — each additional order a customer places is worth significantly more than the last. Even moving a small fraction of single-order customers to a second order changes their expected lifetime value trajectory substantially.

Insight 6: Sao Paulo, Rio de Janeiro, and Minas Gerais represent 66.7% of all orders.

These three states are the platform's commercial core. Any retention programme deployed nationally can be piloted most efficiently in SP, RJ, and MG first given the volume concentration and relatively strong logistics infrastructure.

Insight 7: Northern Brazil faces structural logistics disadvantages.

States in the North (AP, RR, AM) have average delivery times of 24-28 days versus the national average of 12 days. While logistics improvements are out of scope for this project, these states represent a separate retention problem driven by logistics friction and should be treated as a distinct customer segment in any national programme.

Insight 8: Health & Beauty and Watches & Gifts are the highest-revenue categories but not naturally repeat-purchase categories.

Health & Beauty (R$ 1.32M) and Watches & Gifts (R$ 1.20M) lead in revenue but are high-consideration purchase categories with naturally long inter-purchase cycles. This explains part of the structural churn — customers are not failing to return because of a bad experience, but because the category does not naturally generate frequent re-purchase behaviour.

Insight 9: No single variable is a strong independent predictor of churn.

Logistic regression and effect size analysis confirm that delivery delay, review score, price, and payment method are all statistically associated with churn but have negligible effect sizes. The root cause of churn is absence of re-engagement infrastructure — not any single operational failure. This distinguishes Olist's problem from typical retail churn scenarios.

Insight 10: Customers who leave reviews — even negative ones — show marginally higher retention.

The No Review segment has the lowest churn rate (93.75%) when measured at the customer level, suggesting that any post-purchase interaction — including a negative review — correlates with slightly higher likelihood of return. This points to the value of review solicitation and post-purchase engagement tools even when they surface negative feedback.

11. Business Recommendations

Each recommendation follows the format: Insight → Recommendation → Expected Business Impact. Recommendations are prioritised by impact-to-effort ratio.

Recommendation 1 — Launch a Post-Purchase Voucher Re-engagement Programme

> *Insight: Voucher users have a 5.86% repeat rate — double the platform average of 2.81%*

Recommendation: Deploy an automated post-purchase email campaign offering a time-limited voucher (e.g. 10-15% discount) to all first-time customers within 7 days of delivery. Target spend: R$ 15–20 per voucher at scale.

Expected Impact: If the programme converts an additional 2% of one-time customers to repeat buyers (from 2.81% to 4.81%), this would generate approximately 1,900 additional repeat customers from the existing dataset cohort, contributing an estimated R$ 325,000 in incremental revenue at the observed repeat-customer average spend of R$ 170.84 per order.

Recommendation 2 — Build a High-Value Customer Loyalty Tier

> *Insight: High-spend customers churn at 91.78% vs 97.26% for low-spend — 5.5 percentage points lower*

Recommendation: Identify all customers spending above R$ 500 in their first order and enrol them in a named loyalty tier with benefits such as free express shipping, dedicated support, and early access to new categories. The cost per enrolled customer is low; the marginal revenue from retaining even one high-spend customer outweighs it.

Expected Impact: A 10% improvement in high-spend customer retention would add approximately 200 customers to the repeat base, contributing an estimated R$ 170,000 given their above-average order value.

Recommendation 3 — Implement Automated Review-Triggered Retention Workflows

> *Insight: 1-star and 2-star reviews are a leading indicator of disengagement*

Recommendation: Build an automated trigger — when a customer submits a 1-star or 2-star review, initiate a personalised recovery contact within 48 hours: a personal apology from the Olist customer success team, a R$ 20 credit, and a request to understand the issue. This converts a complaint into a relationship signal.

Expected Impact: If 15% of low-review customers who are contacted convert to a second purchase, this addresses approximately 2,400 at-risk customers in the dataset cohort, potentially recovering R$ 410,000 in revenue that would otherwise be lost permanently.

Recommendation 4 — Category-Specific Re-purchase Nudges

> *Insight: Health & Beauty is the top revenue category but has a naturally long re-purchase cycle*

Recommendation: For Health & Beauty, Bed & Bath, and Housewares purchases — categories with consumable or replenishable characteristics — implement a smart reminder programme 60-90 days after first purchase, prompting reorder of the same or related products.

Expected Impact: Even a 3% conversion rate on these reminder campaigns across the top 3 categories would generate approximately 950 additional orders per cohort year, contributing R$ 132,000 at average order value.

Recommendation 5 — Pilot Retention Programmes in SP, RJ, MG First

> *Insight: Three states represent 66.7% of all orders and have the strongest logistics infrastructure*

Recommendation: All retention programmes should be piloted in Sao Paulo, Rio de Janeiro, and Minas Gerais before national rollout. These states provide sufficient volume for statistical significance within 4-6 weeks, have shorter feedback cycles due to better logistics, and represent the highest absolute revenue impact.

Expected Impact: A focused pilot in SP alone (42.2% of orders) would produce results within 6 weeks with sufficient statistical power ($n > 18,000$) to validate or reject each programme's effectiveness before national scaling.

12. Impact Estimation

The following table summarises the estimated revenue impact of all 5 recommendations if implemented simultaneously. Assumptions are conservative (lower-bound estimates) and based on observed data from the 105,000-row dataset.

Recommendation	Target Segment	Assumed Conversion Lift	Est. Additional Customers	Est. Revenue Impact (R$)
Post-purchase voucher programme	All first-time customers	+2% repeat rate	~1,900	~R$ 325,000
High-value loyalty tier	Top spend segment	10% churn reduction	~200	~R$ 170,000
Review-triggered recovery	1-2 star reviewers	15% recovery rate	~2,400	~R$ 410,000
Category re-purchase nudges	Top 3 categories	3% reminder conversion	~950	~R$ 132,000
SP/RJ/MG pilot acceleration	Programme-wide	Faster learning cycle	—	Unlocks full national scale

Combined estimated impact: Approximately R$ 1.04M in recoverable incremental revenue from a customer cohort of 103,000+ unique customers — representing approximately 7.1% of the total dataset revenue of R$ 14.69M. At a national programme scale covering Olist's full customer acquisition volume, the impact would scale proportionally.

Cost of inaction: At the current churn rate of 97.19%, Olist is spending approximately 5-7x more to acquire customers than it would need to if a basic retention infrastructure were in place. Each percentage point improvement in the repeat rate compounds — customers retained in year 1 become the highest-converting audience for year 2 campaigns.

13. Limitations

- Observation window bias: The dataset covers 23 months (Sep 2016 to Aug 2018). Some customers labelled as churned may have intended to return but the observation window closed before their second purchase. The true repeat rate may be slightly higher than 2.81% when observed over a longer period.
- No customer demographics: Age, gender, income level, and urban/rural classification are absent from the dataset. These are significant predictors of re-purchase behaviour in retail and their absence constrains the depth of segmentation possible.
- No marketing exposure data: The dataset contains no information about which customers received any form of re-engagement communication, promotional offer, or loyalty programme. It is not possible to establish causality between any specific intervention and retention outcomes from this data alone.
- Seller quality heterogeneity: Since Olist is a marketplace, product quality varies significantly across sellers. A customer who churned after a bad seller experience is behaviourally different from one who churned for lack of re-engagement stimulus — but the dataset does not distinguish between these two scenarios.
- Geographic representation: SP accounts for 42.2% of observations. Statistical findings reflect SP demographics and logistics disproportionately. Northern and northeastern states are underrepresented and may require separately calibrated interventions.
- Churn definition simplicity: The binary churn definition (1 order = churned) is operationally clear but does not capture purchasing frequency within the retained segment. A customer with 2 orders over 2 years and one with 12 orders over 2 years are both classified as retained.

14. Future Scope

- Predictive churn model: A supervised machine learning model (Random Forest, Gradient Boosting) trained on the features developed in this project could generate individual-level churn probability scores, enabling targeted outreach prioritised by risk score. The logistic regression in NB04 provides a baseline; tree-based models would likely improve performance given the non-linear relationships observed.
- Real-time dashboard: Integration of Olist's live order feed into a Tableau Server or Power BI workspace would convert this static historical dashboard into an operational monitoring tool. Alert thresholds on delivery delay and review score could trigger automated intervention workflows.
- Seller-level analysis: Disaggregating churn by seller would identify which specific sellers are contributing most to customer disengagement. High-churn sellers could be subject to targeted support or listing restrictions.
- Survival analysis: Time-to-second-purchase modelling using Kaplan-Meier curves or Cox proportional hazards regression would provide richer insight into when customers are most likely to return if they are going to, and when the window for re-engagement closes.
- Customer sentiment NLP: The review_comment_message field contains text for 41.3% of orders. Sentiment analysis and topic modelling on this text could surface specific product or service complaints that map to churn, enabling more targeted operational fixes.
- A/B testing framework: The re-engagement programme recommendations in this report require A/B testing to validate effectiveness. A holdout group design across the SP/RJ/MG pilot would generate causal evidence for programme ROI within 6-8 weeks.

15. Conclusion

Olist faces a structural customer retention crisis: 97.19% of customers place exactly one order and never return. This analysis of 105,000 delivered orders across 27 Brazilian states, covering September 2016 to August 2018, confirms that this churn rate is not primarily driven by logistics failures, poor product quality, or regional disadvantage. Olist's delivery performance is excellent — 92.9% of orders arrive ahead of schedule. Customer satisfaction as measured by review scores is strong at 4.07 out of 5. The problem is structural absence of post-purchase engagement infrastructure.

The analysis identifies voucher-based incentives, high-value customer loyalty programmes, review-triggered recovery workflows, and category-specific re-purchase nudges as the four highest-impact retention levers. Combined, conservative implementation of these programmes is estimated to recover approximately R$ 1.04M in incremental revenue from the existing customer cohort — approximately 7.1% of total platform revenue — while building the retention infrastructure needed to improve the repeat rate from 2.81% to a more commercially sustainable 5-8%.

The most important operational recommendation is to start immediately, start in SP/RJ/MG, and start with vouchers — the data is unambiguous that promotional instruments are the single most effective retention mechanism already operating on the platform.

Appendix — Data Dictionary

The following table documents all 19 columns in olist_churn_master.csv (the processed analytical file). The full data_dictionary.md is committed to docs/ in the GitHub repository.

Column	Type	Description	Cleaning Notes
order_id	string	Unique order identifier	No transformation
customer_unique_id	string	Canonical real-customer ID. Use this — not customer_id — for all customer-level analysis	Sourced from df_customers
customer_state	string	Brazilian state code (e.g. SP)	Whitespace stripped
customer_city	string	Brazilian city name	Whitespace stripped
order_purchase_timestamp	datetime	Date and time of order placement	Cast from string via pd.to_datetime
order_delivered_customer_date	datetime	Actual delivery date	Cast; rows with null dropped
order_estimated_delivery_date	datetime	Estimated delivery date shown to customer	Cast from string
order_status	string	Always 'delivered' in this dataset	Filtered to delivered only in NB02
price	float	Item price in BRL	No transformation
freight_value	float	Shipping cost in BRL	No transformation
order_revenue	float	Engineered: price + freight_value	Derived column
payment_type	string	Payment method (credit_card, boleto, voucher, debit_card)	Whitespace stripped; nulls filled 'unknown' in NB05
payment_installments	float	Number of payment installments	Nulls filled with median in NB05
payment_value	float	Total payment value in BRL	Nulls filled with median in NB05
review_score	float	Customer review score 1-5. Null where no review submitted	772 nulls filled with median in NB05
review_comment_message	string	Optional review text. 'no_comment' where absent	Nulls filled 'no_comment'
product_category_name_english	string	English product category name	Translated from Portuguese via category map

Column	Type	Description	Cleaning Notes
delivery_delay_days	int	Engineered: actual minus estimated delivery in days. Negative = early	Derived column
churn	int	Target: 1 = one order (churned), 0 = two or more orders (retained)	Derived at customer_unique_id level

Contribution Matrix

This section documents each team member's contribution across all project phases. Claims are verifiable through GitHub Insights, PR history, and committed files in the repository.

Team Member	Enrol No.	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Praneeth Nakkina Lakshmi	2401010288	—	Minor	Minor	Major	—	—	—
Himank Kaushik	2401010187	Major	—	—	—	—	—	Minor
Priyal Sarda	2401010354	Minor	—	Major (EDA)	—	—	—	—
Gokul VKS	2401020094	—	Major	—	—	Minor	—	Major
Atharva Sharma	2401010112	—	—	—	—	Major	Minor	—
Abhijeet Sinha	2401010013	—	—	—	—	—	Major	—

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead Signature: Praneeth Nakkina Lakshmi Date: 28 April 2026